Riding the Wave?

Who Starts, Follows and Gains from Machine Learning Research Trends

A Study of 54,579 ICLR Submissions, 2017–2026

Minghan Li

Supervisor: A/Prof Jianzhong Qi

Outline

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Motivation: Research Fashions in Machine Learning

Share of ICLR submissions that ...	2017	2026
propose a new architecture	24%	8%
build on a foundation model	0.3%	54%
carry a branded title ("NAME: subtitle")	9%	41%

Are these fashions? What do they displace, who drives them, and are followers rewarded?

Research Questions and Hypotheses

Research question	Key prediction
RQ1 Rise: what does a fashion displace?	More foundation-model (FM) papers in a topic $\rightarrow$ fewer new-architecture papers (H1d)
RQ2 Following: who introduces new work?	Newcomers imitate: fewer new-method papers, more benchmark papers (H2b, H2c)
RQ3 Reward: are followers rewarded?	Crowded topics $\rightarrow$ lower acceptance (H3a)

All predictions were written down before any test was run (10 hypotheses in total).

Related Work and Positioning

	Prior work	Our angle
RQ1	New paradigms displace old work (Kuhn, 1962); foundation models homogenise methods (Bommasani et al., 2021)	Do new architectures fade where foundation models spread?
RQ2	Herding: newcomers imitate (Abrahamson, 1991; Bikhchandani et al., 1992) vs. outsiders bring new ideas (Jeppesen & Lakhani, 2010; Azoulay et al., 2019)	Tell the two views apart
RQ3	Novelty is risky and undervalued (Foster et al., 2015; Wang et al., 2017); reviewers score novel proposals lower (Boudreau et al., 2016)	Is following the fashion rewarded at review?

Data and Measurement

Construct	Measure
Corpus	Provenance-preserving ICLR corpus from our dated official OpenReview snapshots. Existing datasets informed coverage and source audits but were not pooled (Re^2: Zhang et al., 2025; Louis archive: MlouisBE, n.d.; Paper Copilot: Yang et al., 2025)
Papers	43,480 submissions, 2017–2026 (accepted, rejected and withdrawn); 11,099 held-out papers for the replication
Topic	SPECTER2 embeddings of title + abstract (Singh et al., 2023), clustered into $k = 100$ topics; results agree at $k = 75$ and 150
Contribution type	LLM with a fixed codebook (method, benchmark, empirical, theory, other); blind second pass $\kappa = 0.93$
Newcomer paper	No author has an earlier ICLR submission in the topic (73% of papers)
Acceptance	Final decision; locked away until the single registered run

Hindsight Bias in Topic Maps

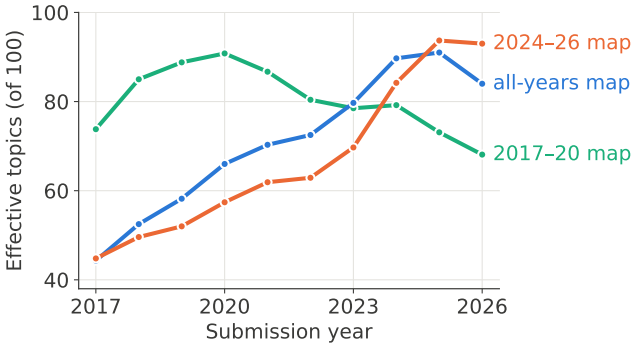


Figure 1: Topic diversity per year under three topic maps.

Topic map

The 100 topic clusters, fitted on papers from chosen years; every paper is then assigned to its nearest topic.

Each map finds peak diversity in the years it was fitted on.

⇒ The fitting window shapes the history one sees.

Consequence: Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** on the 2017–20 map vs. **29** on the all-years map.
- Do topics age, from new methods towards benchmarks? All-years map: no. 2017–20 map: yes.

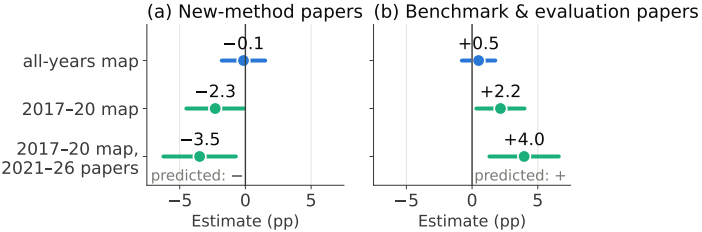


Figure 2: Same papers, same model: coefficient on topic maturity, 95% CI (2017–20 rows exploratory).

$\Rightarrow$ Every registered result must also hold on the 2017–20 map and a year-balanced map.

Empirical Strategy

Model (linear probability)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{\text{topic}(i)} \;+\; \tau_{\text{year}(i)} \;+\; \varepsilon_i$$

- y_i: paper type or acceptance (0/1); x_i: e.g. newcomer paper
- Papers compared within the same topic and year; standard errors clustered by topic

Supported only if

1. predicted sign, Holm-adjusted $p < 0.05$
2. same sign in $\geq 75\%$ of alternative specifications
3. same sign on both hindsight checks

Protocol

Registered in advance, acceptance locked, code frozen, run once; held-out tests registered before their run.

RQ1: Where Foundation Models Spread, New Architectures Fade

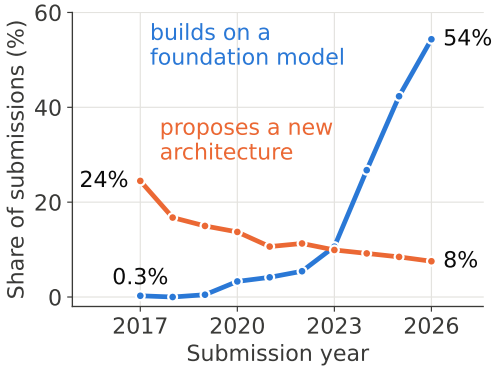


Figure 3: Field-wide trends (descriptive).

Within topic and year, FM adoption 0 → 100%:

New-architecture papers	Discovery	Held-out
Change (pp)	-8.0	-11.0
95% CI	$[-12.1, -3.8]$	$[-18.1, -3.9]$

Supported and replicated

Topics at 50% vs. 10% FM adoption: about one third fewer new-architecture papers.

RQ2: Newcomers Bring New Methods, Opposite to Our Prediction

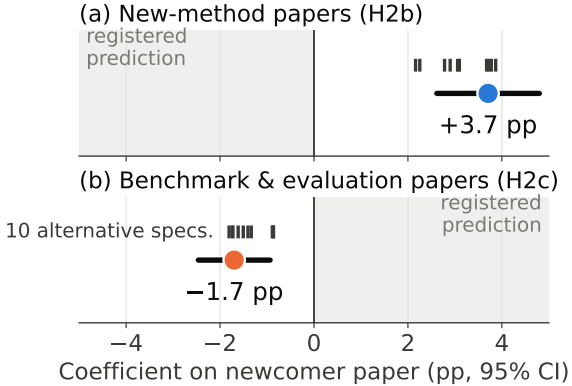


Figure 4: Discovery; within topic and year; grey: predicted side.

Newcomer paper (pp)	Method	Benchmark
Predicted	−	+
Discovery	+3.7	−1.7
Held-out	+3.6	−2.1

Reversed predictions registered, then **replicated** on held-out papers.

Reading

Consistent with outsider innovation, not with herding.

RQ3: Is Following the Fashion Rewarded?

Association with acceptance (pp, 95% CI)	Discovery	Held-out
Crowded topic, per log unit of prior papers (H3a)	-1.7 $[-3.7, +0.3]$	-1.5 $[-5.2, +2.2]$
Branded title (exploratory)	no premium	–
New architecture $\times$ topic FM adoption (exploratory)	$+8.3$ $[+1.3, +15.4]$	–

Table 1: Within topic and year. Crowding: neither split could detect effects below about 4–5 pp.

Key point

No detectable reward for following the fashion; papers that go against the tide may even do better (exploratory).

Do the Results Hold on Held-out Papers?

Three of four key results replicated; crowding stays inconclusive.

Hypothesis	Pred.	Discovery	Held-out [95% CI]	Verdict
H1d FM adoption → new-architecture papers	−	−8.0	−11.0 [−18.1, −3.9]	supported
H2b′ newcomer → new-method paper	+	+3.7	+3.6 [+1.5, +5.7]	supported
H2c′ newcomer → benchmark paper	−	−1.7	−2.1 [−3.3, −0.8]	supported
H3a crowded topic → acceptance	−	−1.7	−1.5 [−5.2, +2.2]	inconclusive†

Table 2: 11,099 papers from a released former test split, never analysed before one registered run: an out-of-sample replication, not a certified untouched holdout. Supported: one-sided Holm $p < 0.05$; pp. † Too weak to tell (below MDE_{80}).

Also registered: post hoc tests from discovery, e.g. far jumps from a team's own past work go with lower acceptance (replicated). Full table in the report.

Contributions

1. **Method:** topic maps fitted with hindsight can manufacture trends; an early-year map is a cheap check.
2. **Design:** registered, outcome-blind, single-run protocol, with a held-out replication.
3. **Findings** (associations):
   - where foundation models spread, new architectures fade (replicated)
   - newcomers lean towards new methods, not imitation (replicated)
   - no detectable acceptance reward for following the fashion

Limitations and Future Work

Limitations

- One venue; associations, not causes
- LLM labels: consistent ($\kappa = 0.93$); hand-checked only on a small random sample

Future work

- A prospective test on future ICLR submissions
- Larger human validation of the labels
- More semantic labels, e.g. how exaggerated or playful a title is (in progress)

Thank you

Questions and comments welcome

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References

Abrahamson, E. (1991). Managerial fads and fashions: The diffusion and rejection of innovations. *Academy of Management Review*, 16(3), 586–612.

Azoulay, P., Fons-Rosen, C., & Graff Zivin, J. S. (2019). Does science advance one funeral at a time? *American Economic Review*, 109(8), 2889–2920.

Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026.

Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. *arXiv:2108.07258*.

Boudreau, K. J., Guinan, E. C., Lakhani, K. R., & Riedl, C. (2016). Looking across and looking beyond the knowledge frontier: Intellectual distance, novelty, and resource allocation in science. *Management Science*, 62(10), 2765–2783.

Foster, J. G., Rzhetsky, A., & Evans, J. A. (2015). Tradition and innovation in scientists' research strategies. *American Sociological Review*, 80(5), 875–908.

Jeppesen, L. B., & Lakhani, K. R. (2010). Marginality and problem-solving effectiveness in broadcast search. *Organization Science*, 21(5), 1016–1033.

Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.

MlouisBE. (n.d.). *iclr-rebuttal-analysis* [ICLR archive; GitHub repository]. github.com/MlouisBE/iclr-rebuttal-analysis.

Singh, A., D'Arcy, M., Cohan, A., Downey, D., & Feldman, S. (2023). SciRepEval: A multi-format benchmark for scientific document representations. In *Proceedings of EMNLP*.

Wang, J., Veugelers, R., & Stephan, P. (2017). Bias against novelty in science: A cautionary tale for users of bibliometric indicators. *Research Policy*, 46(8), 1416–1436.

Yang, J., Wei, Q., & Pei, J. (2025). Paper Copilot: Tracking the evolution of peer review in AI conferences. *arXiv:2510.13201*.

Zhang, D., Bao, Z., Du, S., Zhao, Z., Zhang, K., Bao, D., & Yang, Y. (2025). Re^2: A consistency-ensured dataset for full-stage peer review and multi-turn rebuttal discussions. *arXiv:2505.07920*.